

In 1999, the GSM standard was modified to include EDGE (enhanced data rates for GSM evolution). By 2003, providers had started implementing EDGE. EDGE is a technology backward compatible to GPRS. It allows for transfer rates as high as 1 Mbps.

With these improvements on GSM and equivalent ones in CDMA, mobile technology continues to improve. Today, both GSM and CDMA are threatened by 3G services such as UMTS. Steps have been taken towards even the next generation, namely 4G. Let us take a look at these.

4.5 Emerging technologies

Most of the emerging technologies in the field of cellular telephony can be classified under 3G. Standing for third generation, this is the name given to a class of technologies that involve GSM with EDGE, UMTS (a new 3G mobile standard) and CDMA2000, an advanced version of the IS 95 CDMA standard. 3G also includes WiMax (Long range wireless LAN) and DECT (Digitally Enhanced Cordless Telephony, the technology used in modern cordless phones.) The unifying features of these technologies are high data rates, high security, etc.

UMTS

A 3G technology otherwise known as W-CDMA (Wideband Code Division Multiple Access), UMTS combines the features of CDMA and GSM to create a completely new technology. While UMTS phones require completely new networks (completely new cell transmitters) the technology is quite compatible with GSM, and UMTS phones can work on both GSM and CDMA networks with reduced functionality. In India, UMTS services are provided only by BSNL mobile, and only in selected cities. UMTS allows a maximum theoretical data rate of about 21 megabits per second, though deployed capacities currently support only up to seven megabits per second.

UMTS runs on the frequencies of 2,100 and 1,700 MHz. Canada uses 1,900 MHz while other parts of the world can be found to use 850 MHz.

With promotional campaigns focussing on high bandwidth applications such as videoconferencing and mobile TV, UMTS